



UNIVERSITÀ
DEGLI STUDI
FIRENZE

DINFO

Dipartimento di
Ingegneria dell'Informazione



Laboratorio di Elettronica Digitale

Microcontrollers

Enrico Boni





Programmable digital devices

- Microcontrollers (AVR, PIC)
- Microprocessors (ARM, AMD, INTEL)
- Digital signal processors (DSP) (TMS320C6X)
- Graphic Processing Unit (GPU) (NVIDIA Kepler)
- Massively Parallel Processor Arrays (MPPA) (Ambric AM2045)
- Complex Programmable Logic Devices (CPLD) (MAX3000)
- Field Programmable Gate Array (FPGA) (EP4S, EP5GX)

Programmable digital devices

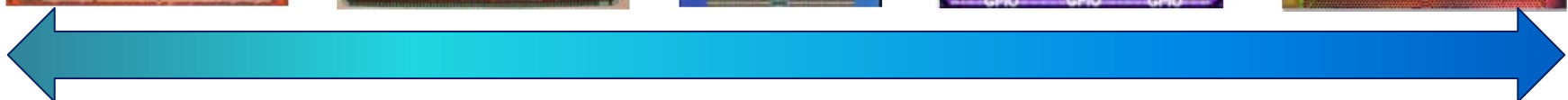
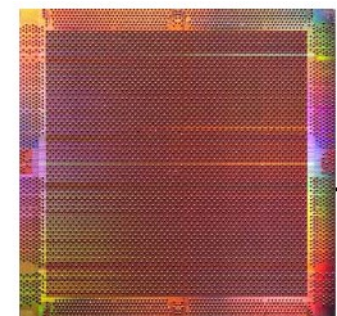
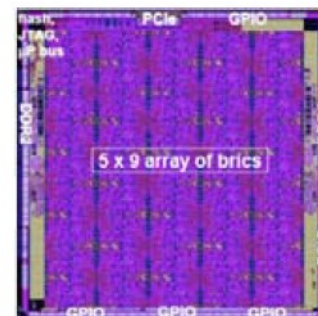
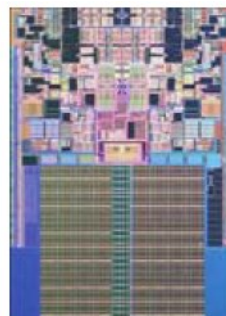
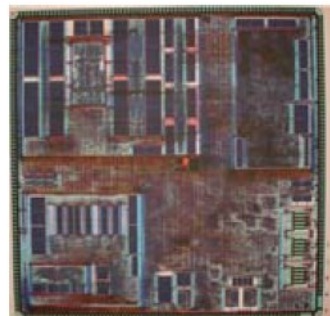
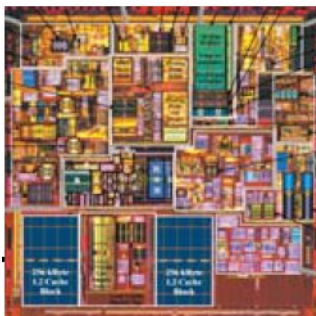
CPU

DSP

Multicores

Arrays

FPGA

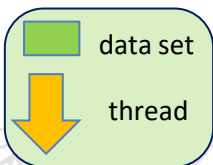


↑
Single core

↑
Multicores Coarse-Grained
CPUs and DSPs

↑
Coarse-Grained
Massively Parallel
Processor Arrays

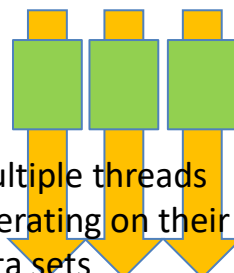
↑
Fine-Grained
Massively Parallel
Arrays



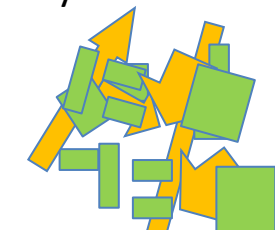
Single thread
operating on its
data set



Multiple threads
operating on their
data sets



Multiple threads
operating on
multiple data sets



'unrelated' threads
and data sets



Definitions

- Microprocessor:
 - Single chip processing unit (e.g. 8086)
 - Needs external circuits to work:
 - RAM
 - DMA Controller (Intel 8237)
 - UART (8250)
 - Interrupt controller (Intel 8259)





Definitions

- **Microcontroller:**
 - Single chip processing unit
 - Usually simpler than a Microprocessor
 - Doesn't need external circuits to work:
 - Embedded RAM, FLASH, DMA, PIC
 - Rich set of embedded peripherals





MC Requirements

- Fast and predictable code execution
 - RISC core
 - CISC core with small instruction set
 - Harvard vs Von Neumann memory architecture





MC Requirements

- Reduced footprint:
 - Small dimensions, integrated system on chip
 - Power-wise electronics (down to few μW)
 - Battery operated devices
 - Miniaturized power supply circuits





MC Requirements

- Easy interface with external world:
 - GPIO, bit addressable
 - Communication peripherals
 - Mixed signal peripherals (ADC/DAC)
 - Numerical Control interfaces (PWM)
 - Fast reaction to external events (HW IRQ)





MC Requirements

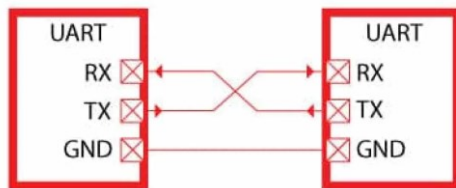
- Communication peripherals:
 - UART, SPI, I2C
 - CAN
 - USB, Ethernet





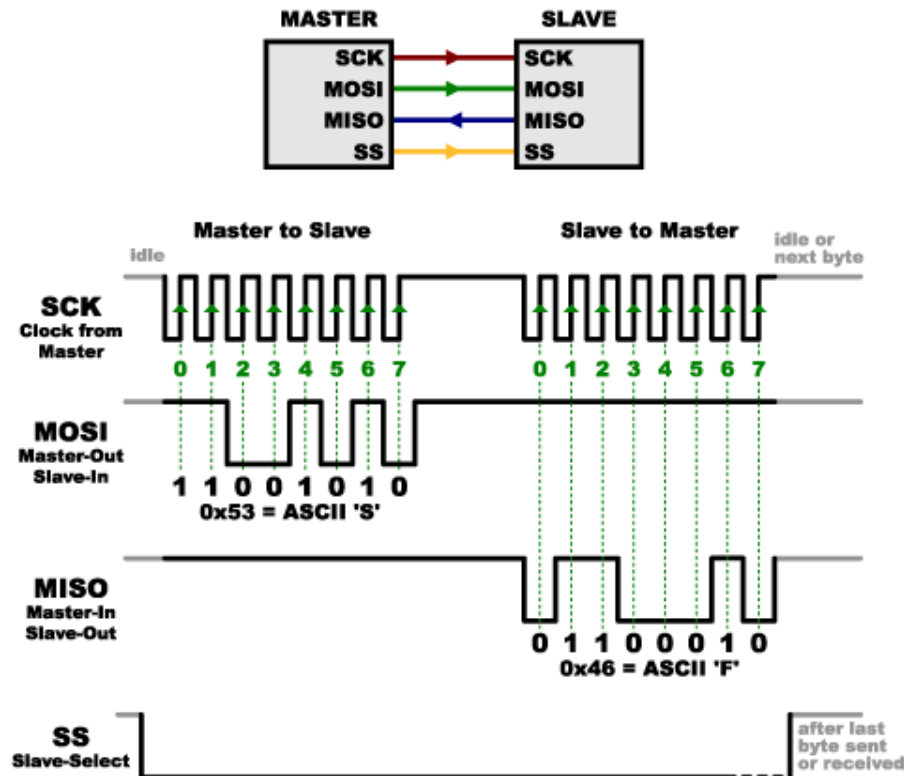
UART:

(Universal Asynchronous Receiver-Transmitter)



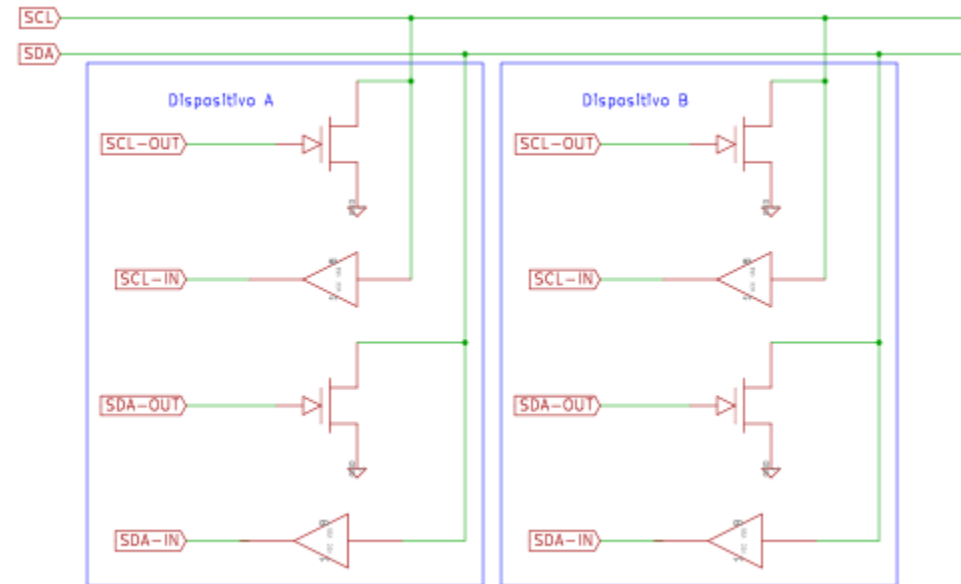
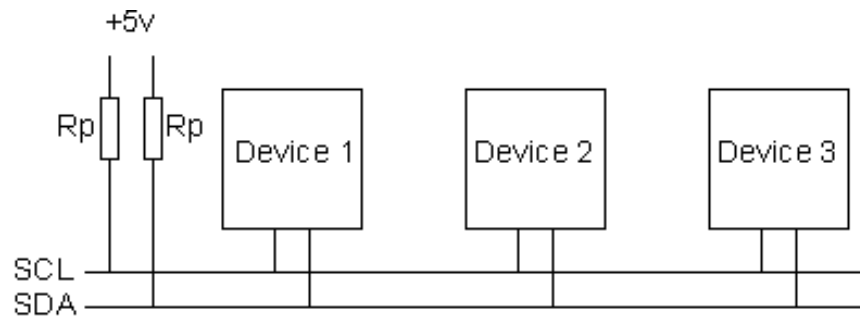
- Asynchronous: Master and slave need to have a 'common' timing source.

SPI: (Serial Peripheral Interface)



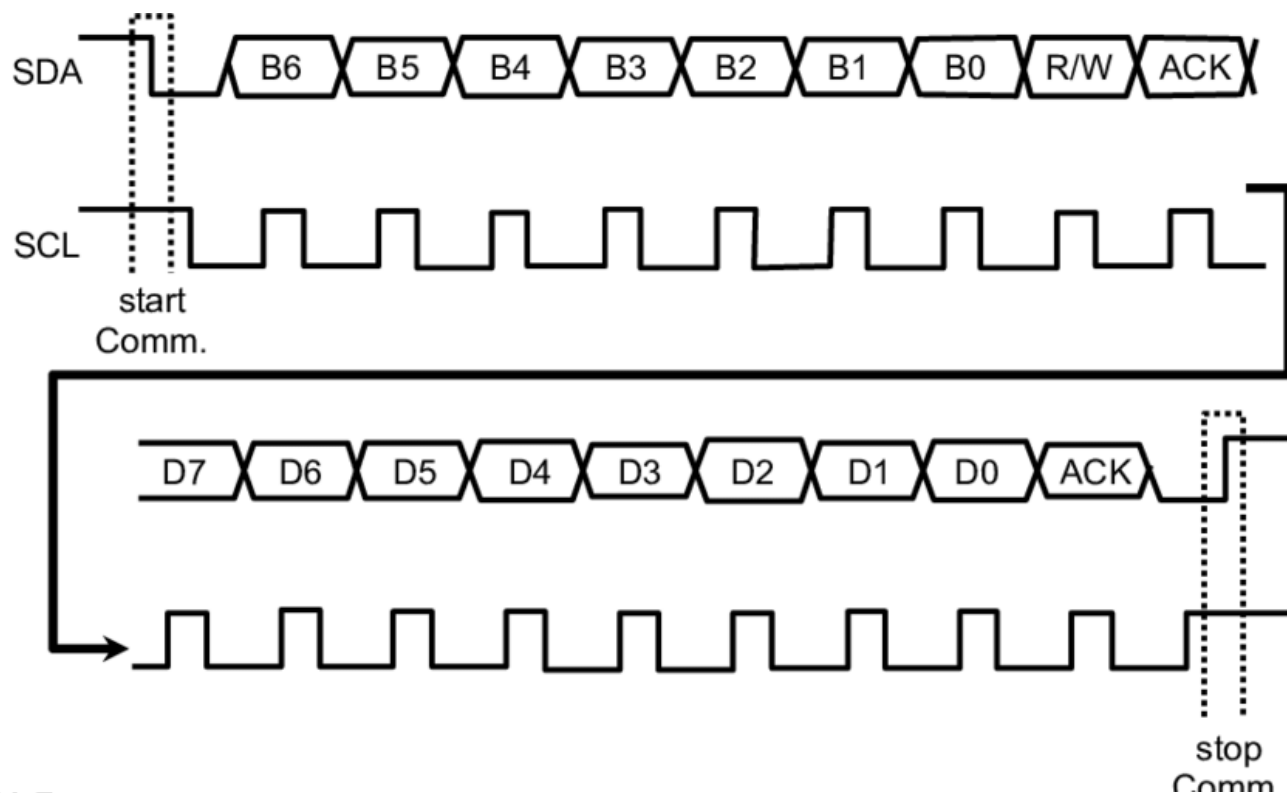
- Synchronous, but more wires.

I2C: (Inter-Integrated Circuit)





I2C: (Inter-Integrated Circuit)



- Synchronous, only 2 wires.



I2C: (Inter-Integrated Circuit)

- Typical transactions:
 - Writing a command
 - Writing data to a memory
 - Reading data (from a memory/sensor)



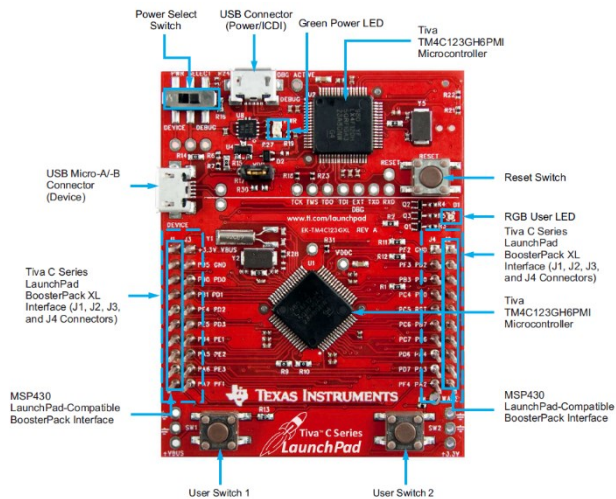
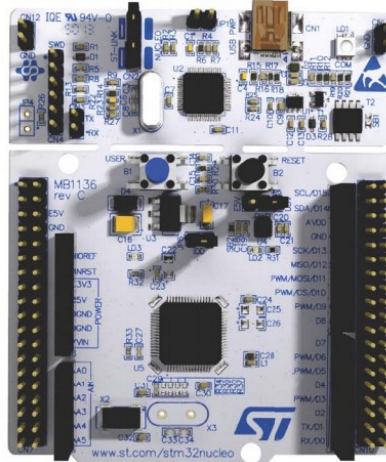


I2C: (Inter-Integrated Circuit)

- Typical pull-up resistors:
 - 100kbit/s $\rightarrow$ 10k Ohm
 - 400kbit/s $\rightarrow$ 2.2k Ohm
 - 1Mbit/s $\rightarrow$ 1k Ohm



Some uC boards

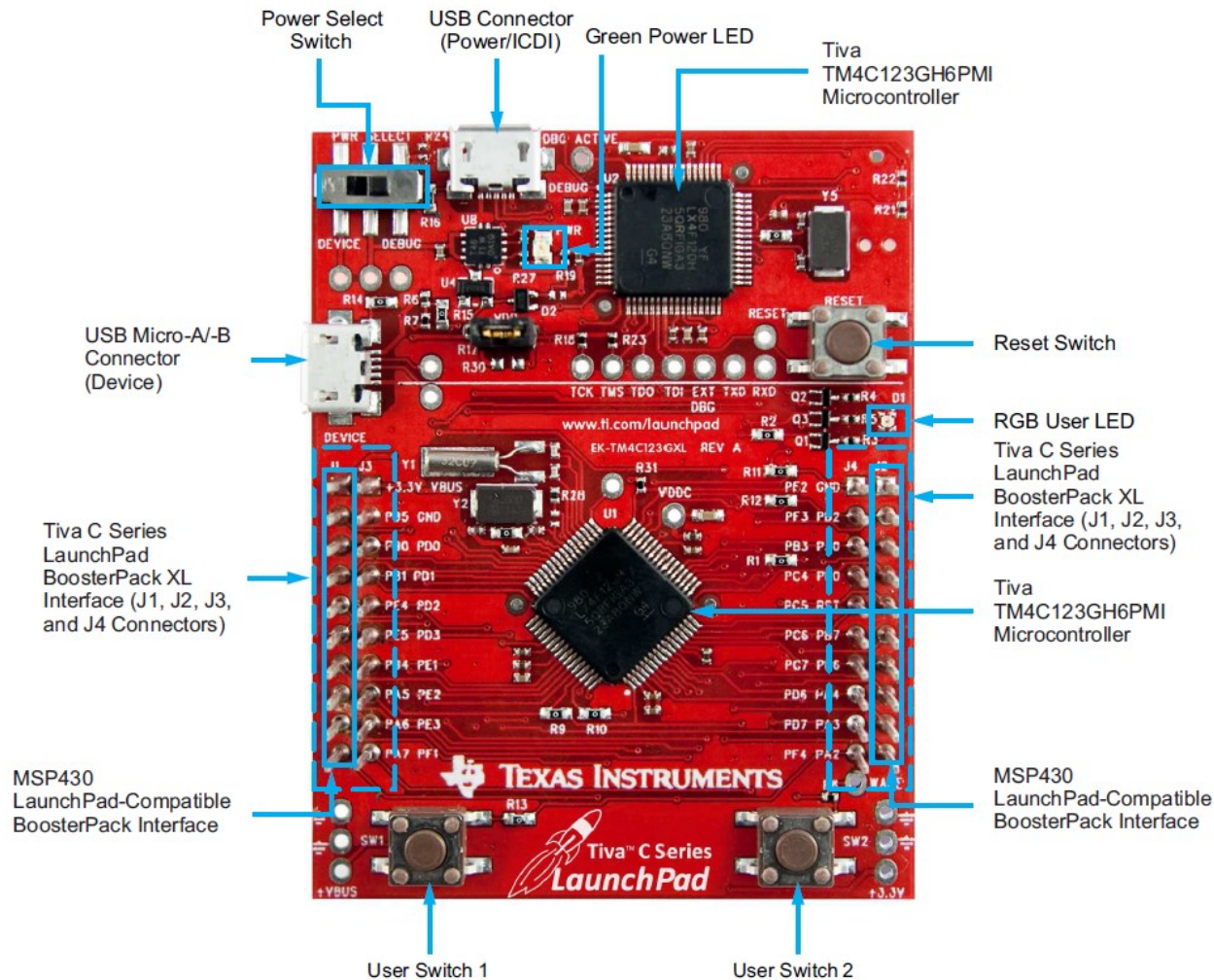




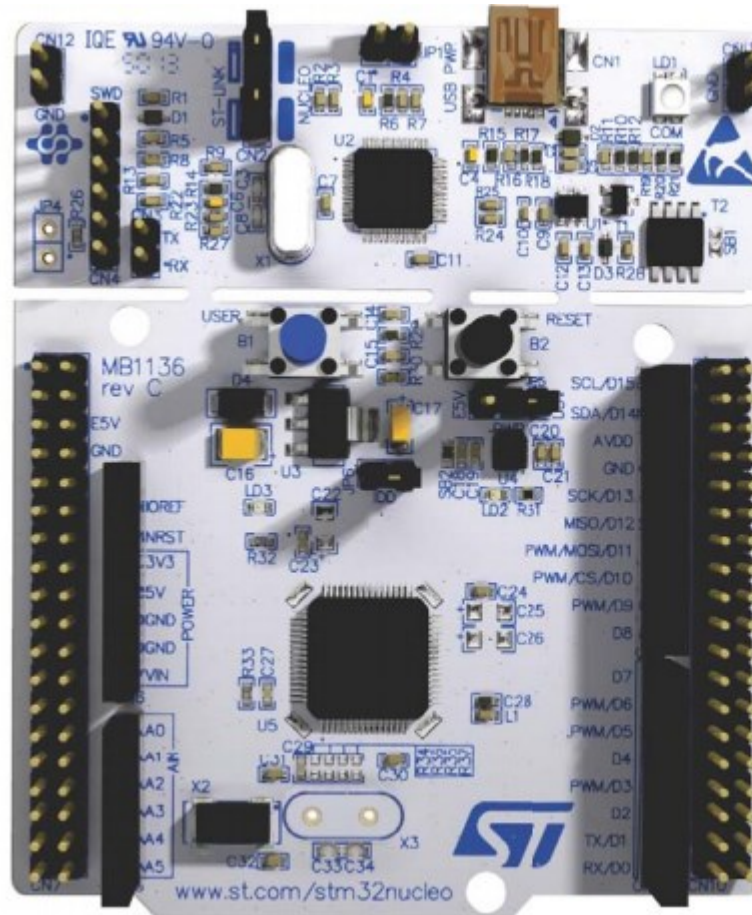
Arduino Board:



Tiva-c board:

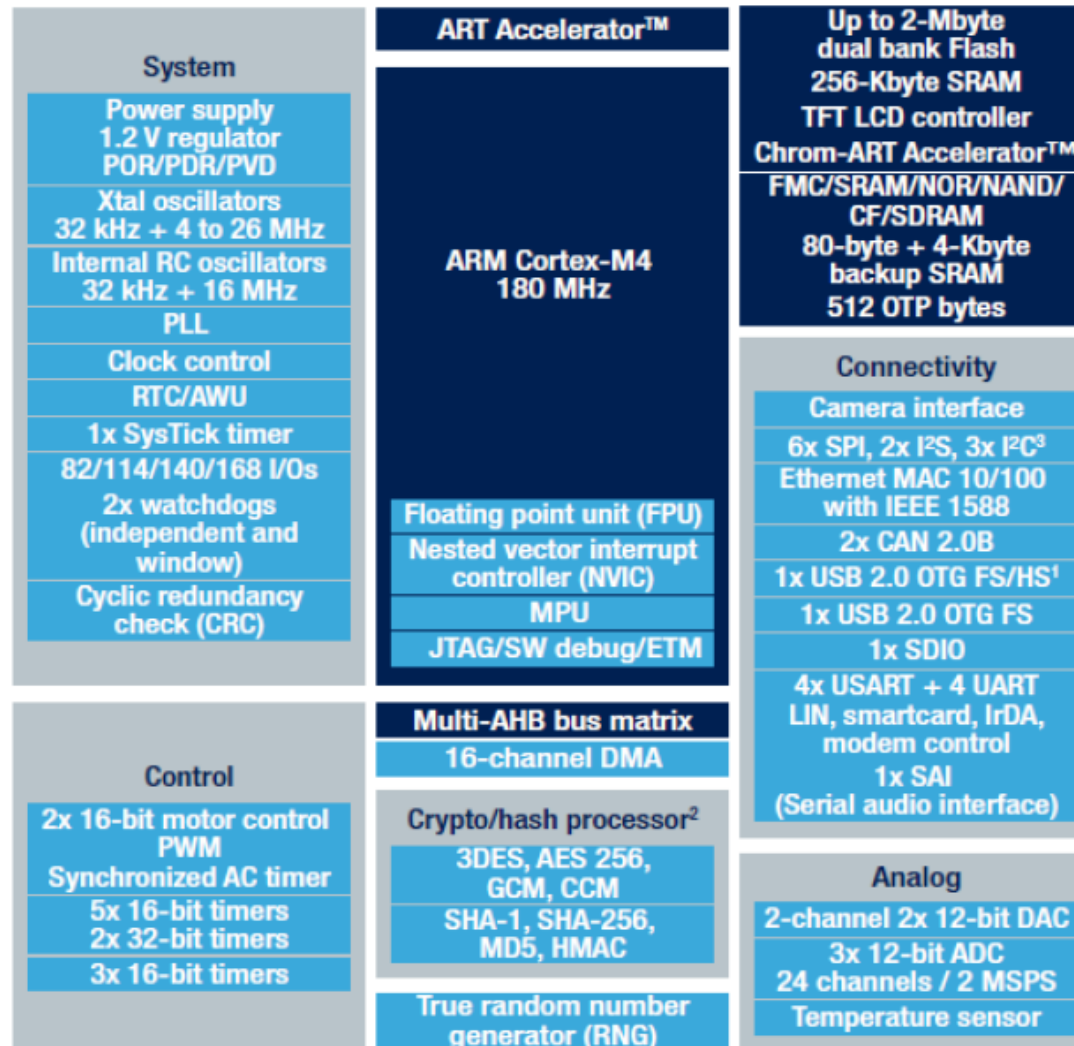


STM32 Nucleo:

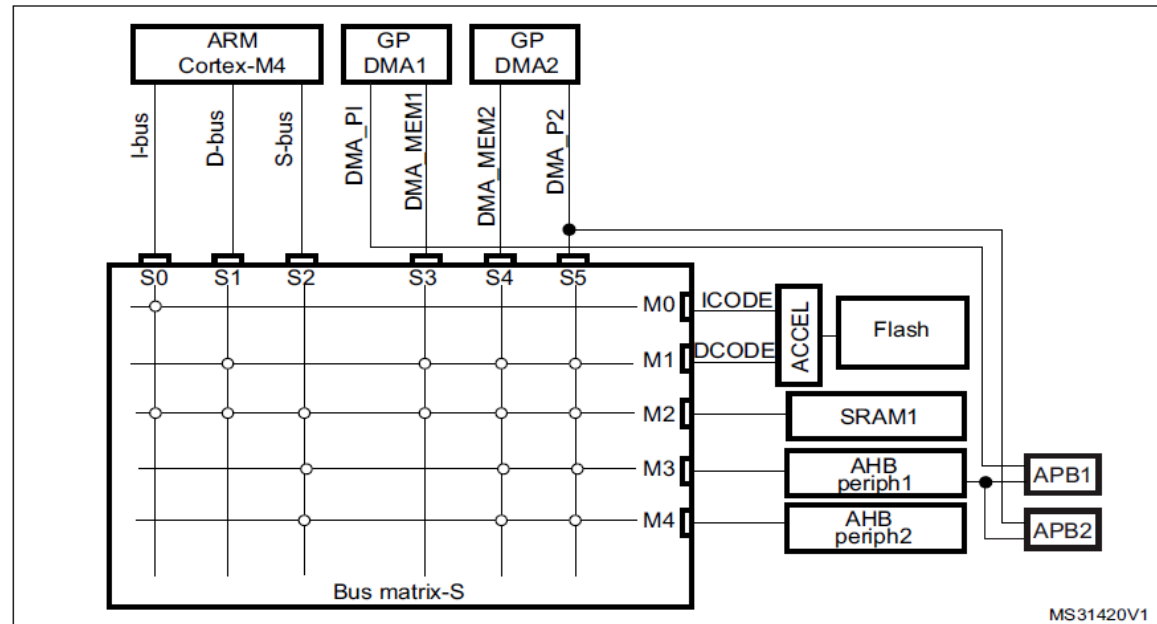




STM32 Cortex-M4 (F4)



Cortex-M4 CPU:

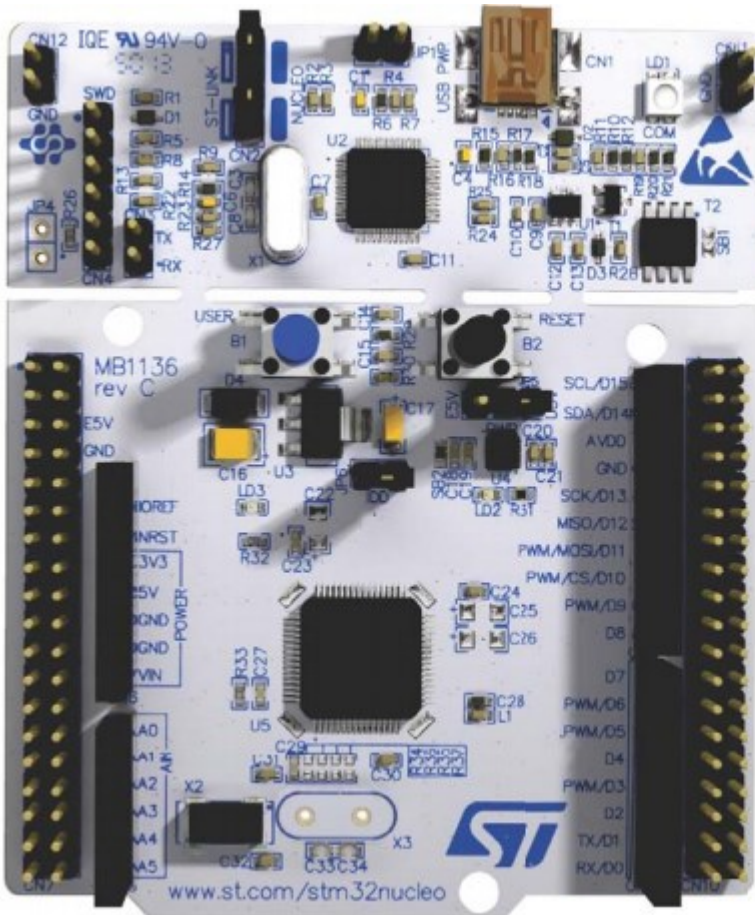


1. STM32F411xC/E: 256 KBytes / 512KBytes Flash with 128 KBytes SRAM

- Harvard architecture
- Floating Point Unit
 - Single precision floating point unit
- Low power (!)
 - 100μA/MHz



Lab evaluation boards:



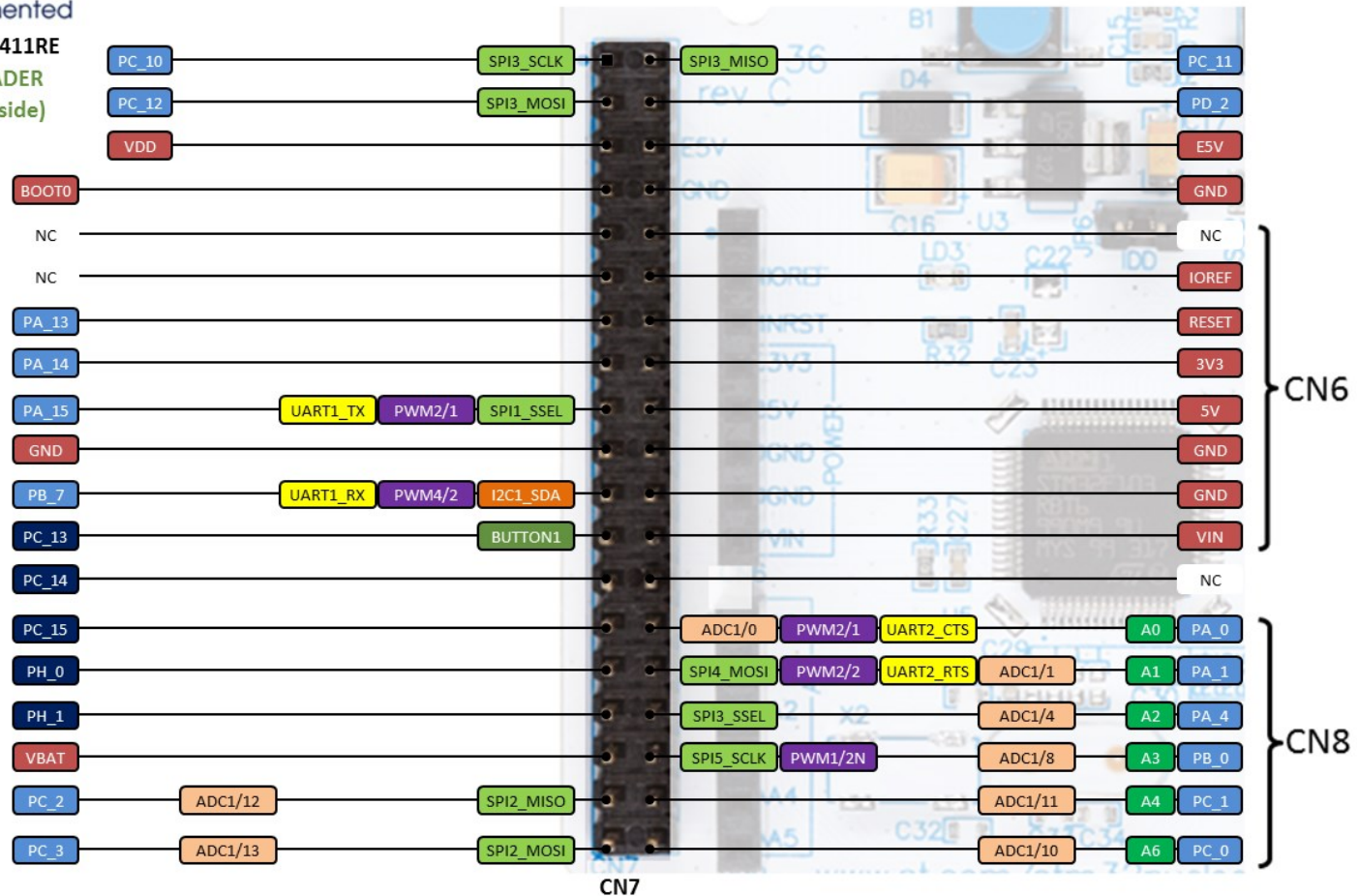
- NUCLEO-F401
- NUCLEO-F411
- NUCLEO-L476



NUCLEO-F411RE

CN7 HEADER

(top left side)



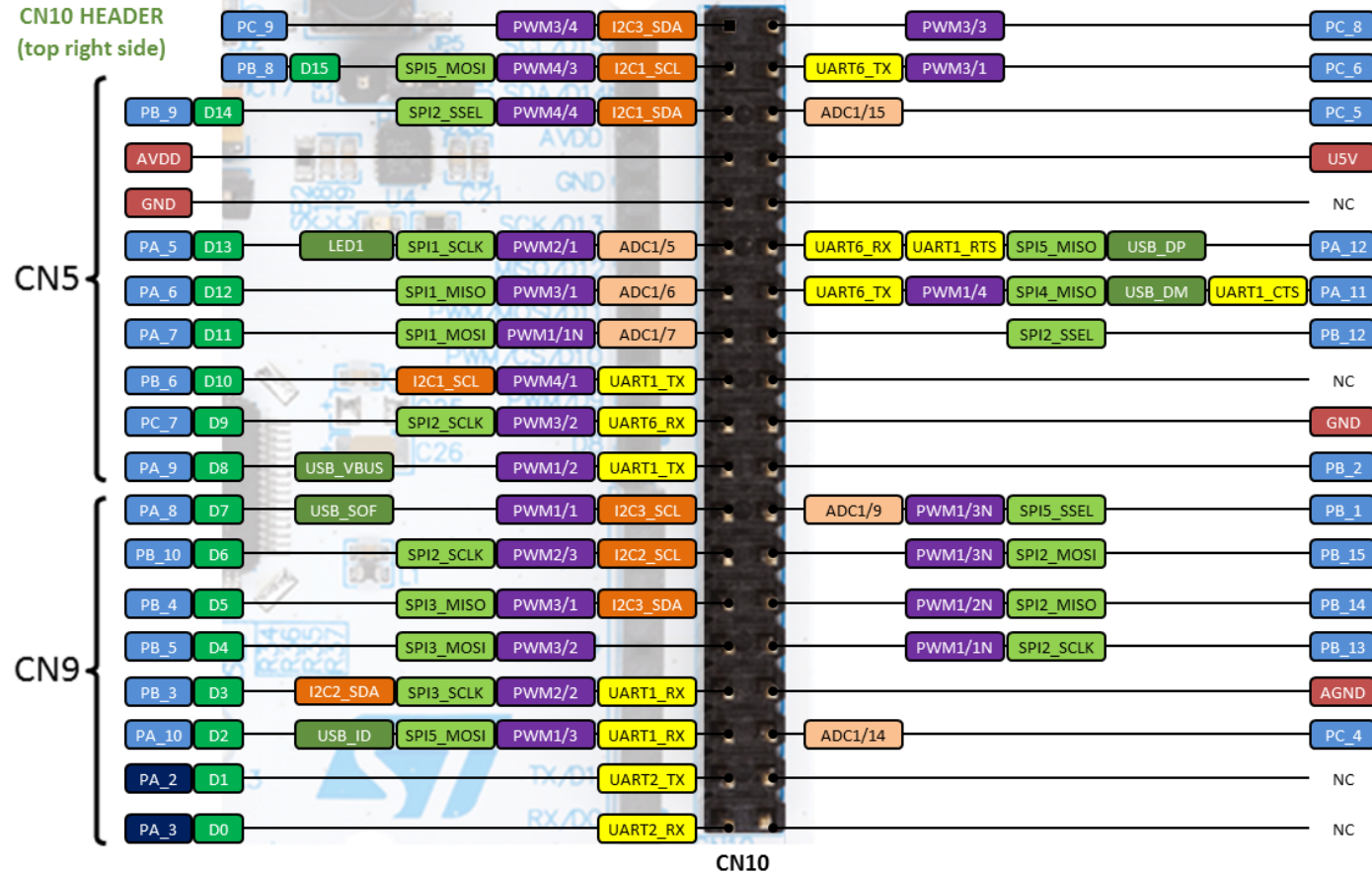


life.augmented

NUCLEO-F411RE

CN10 HEADER

(top right side)



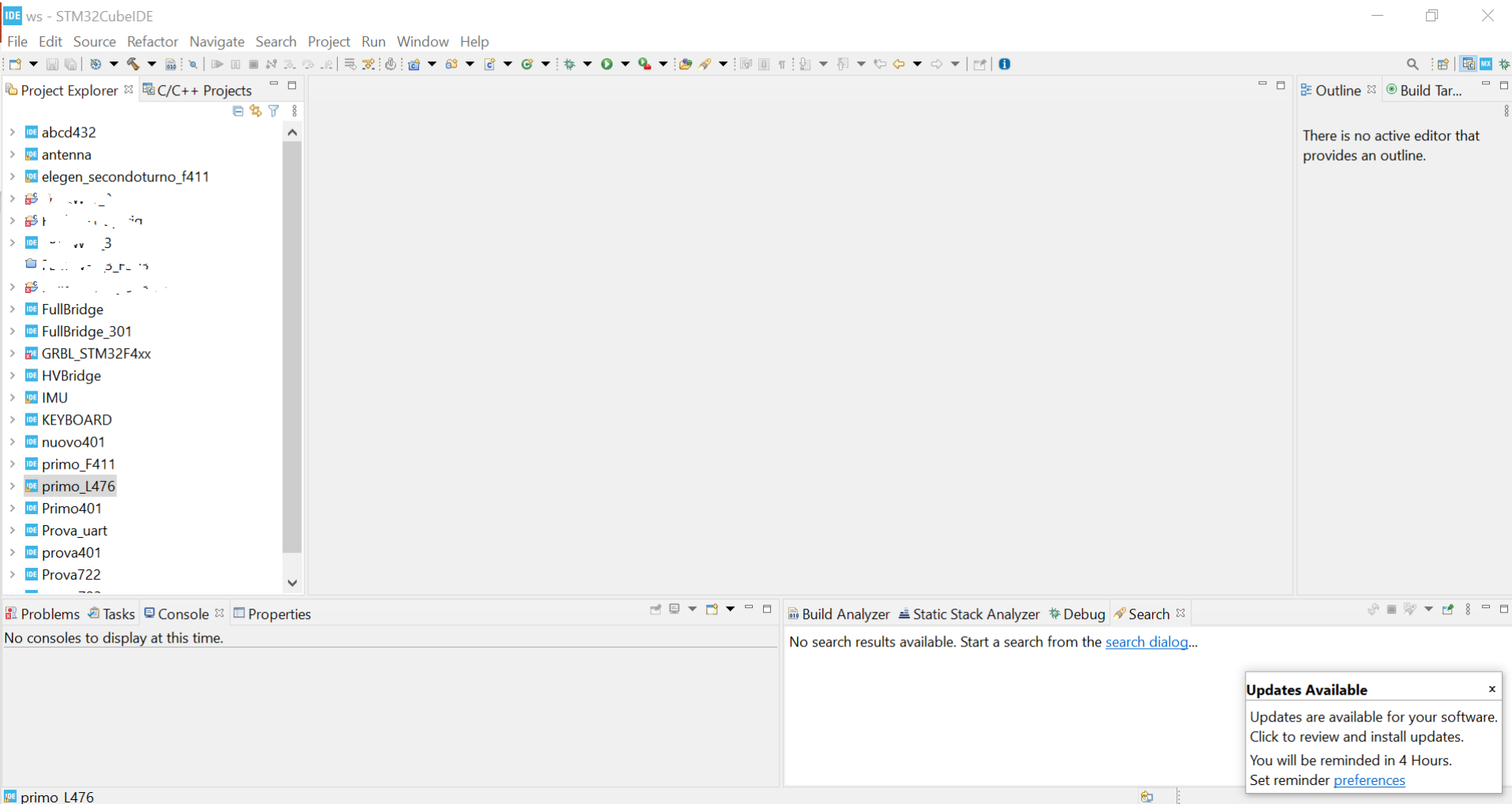


STM32 development tool:

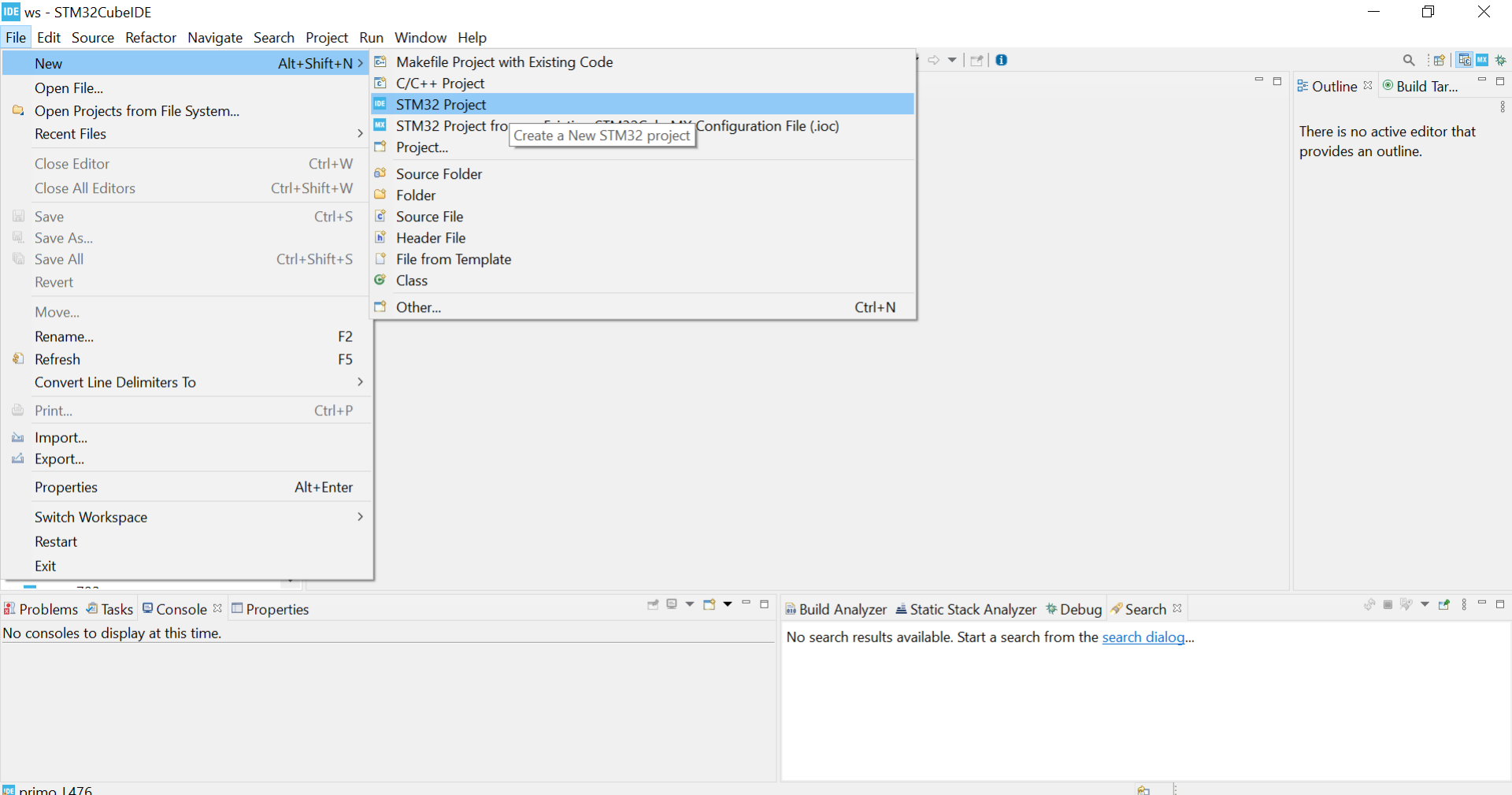
- STM32CubeIDE



STM32CubeIDE



STM32CubeIDE



STM32CubeIDE

STM32 Project

Target Selection

STM32 target or STM32Cube example selection is required

MCU/MPU Selector

Board Selector

Example Selector

Cross Selector

Type

Check/Uncheck All

- ☐ Discovery Kit
- ☐ Evaluation Board
- ☐ Evaluation Kit
- ☐ Nucleo USB Dongle
- ☐ Nucleo-144
- ☒ Nucleo-64
- ☐ Nucleo-144 Kit

MCU/MPU Series

Check/Uncheck All

- ☐ STM32F0
- ☐ STM32F1

Features

Large Picture

Docs & Resources

Datasheet

Buy



ST MCU Finder
All STM32 & STM8
MCUs in one place



Boards List: 9 items

Export

	Overview	Commercial Part...	Type	Marketing Status	Unit Price (US\$)	Mounted Device
☆		NUCLEO-L412RB...	Nucleo-64	Active	15.0	STM32L412RBTxP
☆		NUCLEO-L433RC...	Nucleo-64	Active	15.0	STM32L433RCTxP

< Back

Next >

Finish

Cancel

STM32CubeIDE

STM32 Project

Target Selection

STM32 target or STM32Cube example selection is required

MCU/MPU Selector Board Selector Example Selector Cross Selector

MCU/MPU Series

Check/Uncheck All

☐ STM32F0

☐ STM32F1

☐ STM32F2

☐ STM32F3

☒ STM32F4

☐ STM32F7

☐ STM32G0

☐ STM32G4

☐ STM32H7

☐ STM32L0

☐ STM32L1

☒ STM32L4

☐ STM32L5

Features

Large Picture

Docs & Resources

Datasheet

Buy



Boards List: 9 items

Export

*	Overview	Commercial Part...	Type	Marketing Status	Unit Price (US\$)	Mounted Device
		NUCLEO-L412RB...	Nucleo-64	Active	15.0	STM32L412RBTxP
		NUCLEO-L433RC...	Nucleo-64	Active	15.0	STM32L433RCTxP

< Back

Next >

Finish

Cancel

STM32CubeIDE

IDE STM32 Project

Target Selection

Select STM32 target or STM32Cube example

IDE

MCU/MPU Selector Board Selector Example Selector Cross Selector

- ☐ STM32F0
- ☐ STM32F1
- ☐ STM32F2
- ☐ STM32F3
- ☒ STM32F4
- ☐ STM32F7
- ☐ STM32G0
- ☐ STM32G4
- ☐ STM32H7
- ☐ STM32L0
- ☐ STM32L1
- ☒ STM32L4
- ☐ STM32L4+
- ☐ STM32L5
- ☐ STM32MP1
- ☐ STM32WB

Features

Large Picture

Docs & Resources

Datasheet

Buy

STM32F4 Series

NUCLEO-F411RE

STMicroelectronics NUCLEO-F411RE Board Support and Examples

Unit Price (US\$) : 13.0

Boards List: 9 items

Export

*	Overview	Commercial Part...	Type	Marketing Status	Unit Price (US\$)	Mounted Device
		NUCLEO-F410RB	Nucleo-64	Active	13.0	STM32F410RBTx
		NUCLEO-F411RE	Nucleo-64	Active	13.0	STM32F411RETx
		NUCLEO-F446RE	Nucleo-64	Active	14.0	STM32F446RETx



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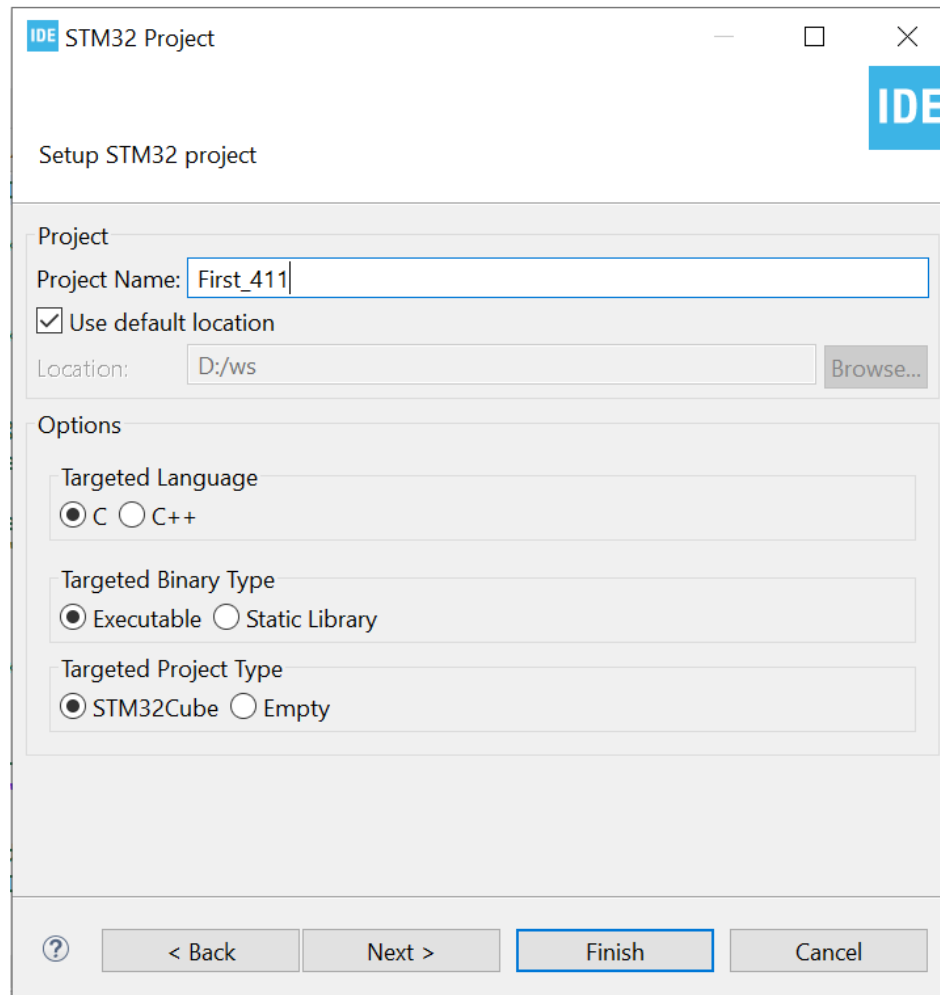
Next >

Finish

Cancel



STM32CubeIDE



IDE STM32 Project

Setup STM32 project

Project

Project Name: First_411

☒ Use default location

Location: D:/ws Browse...

Options

Targeted Language

☒ C ☐ C++

Targeted Binary Type

☒ Executable ☐ Static Library

Targeted Project Type

☒ STM32Cube ☐ Empty

? < Back Next > Finish Cancel

Target Configuration

IDE ws - Device Configuration Tool - STM32CubeIDE

File Edit Navigate Search Project Run Window Help

Project Explorer

- First_411
 - Includes
 - Core
 - Inc
 - Src
 - main.c
 - stm32f4xx_hal_msp.c
 - stm32f4xx_it.c
 - syscalls.c
 - systemem.c
 - system_stm32f4xx.c
 - Startup
 - Drivers
 - First_411.ioc
 - STM32F411RETX_FLASH.ld
 - STM32F411RETX_RAM.ld
 - FullBridge
 - FullBridge_301
 - GRBL_STM32F4xx
 - HVBridge
 - IMU
 - KEYBOARD
 - nuovo401
 - primo_F411
 - primo_L476
 - Primo401
 - Prova_uart
 - prova401
 - Prova722
 - prova723
 - provaL4q

First_411.ioc

Pinout & Configuration

Clock Configuration

Project Manager

Tools

Software Packs

Pinout

Pinout view

System view

Categories A->Z

- System Core
- Analog
- Timers
- Connectivity
- Multimedia
- Computing
- Middleware

STM32F411RETx LQFP64

Pinout view showing the STM32F411RETx LQFP64 package with pins labeled VDD, VSS, BOOT, PB9, PB8, PB7, PB6, PB5, PB4, PB3, PB2, PB1, PB0, PC12, PC11, PC10, PC9, PC8, PC7, PC6, PC5, PC4, PC3, PC2, PC1, PC0, VSSA, VDDA, PA0, PA1, PA2, PA3, PA4, PA5, PA6, PA7, PA8, PA9, PA10, PA11, PA12, PA13, PA14, PA15, PC13, PC14, PC15, PH0, PH1, NRST, T_RX, Led, USART_TX, B1 [Blue PushButton], RCC_OSC32_IN, RCC_OSC32_OUT, RCC_OSC_IN, RCC_OSC_OUT, VBAT, YSS, TMS.

Console Search

No consoles to display at this time.

No consoles to display at this time.

Target Configuration: clock

IDE ws - Device Configuration Tool - STM32CubeIDE

File Edit Navigate Search Project Run Window Help

Project E... First_411.ioc

Pinout & Configuration Clock Configuration Project Manager Tools

Resolve Clock Issues

Input frequency 32.768 KHz

LSE

LSI RC 32 32 KHz

HSI RC 16 16 MHz

Input frequency 8 1-50 MHz

HSE

PLL Source Mux

System Clock Mux

RTC Clock Mux

PLLClock

Enable CSS

Main PLL

SYSCLK (MHz) 84

AHB Prescaler / 1

HCLK (MHz) 84 100 MHz max

APB1 Prescaler / 2

APB2 Prescaler / 1

PCLK1 42 50 MHz max

PCLK2 84 100 MHz max

Ethernet PTP clock (MHz) 84

HCLK to AHB bus, core, memory and DMA (MHz) 84

To Cortex System timer (MHz) 84

FCLK Cortex clock (MHz) 84

APB1 peripheral clocks (MHz) 42

APB1 Timer clocks (MHz) 84

APB2 peripheral clocks (MHz) 84

APB2 timer clocks (MHz) 84

48MHz clocks (MHz) 84

I2S source Mux

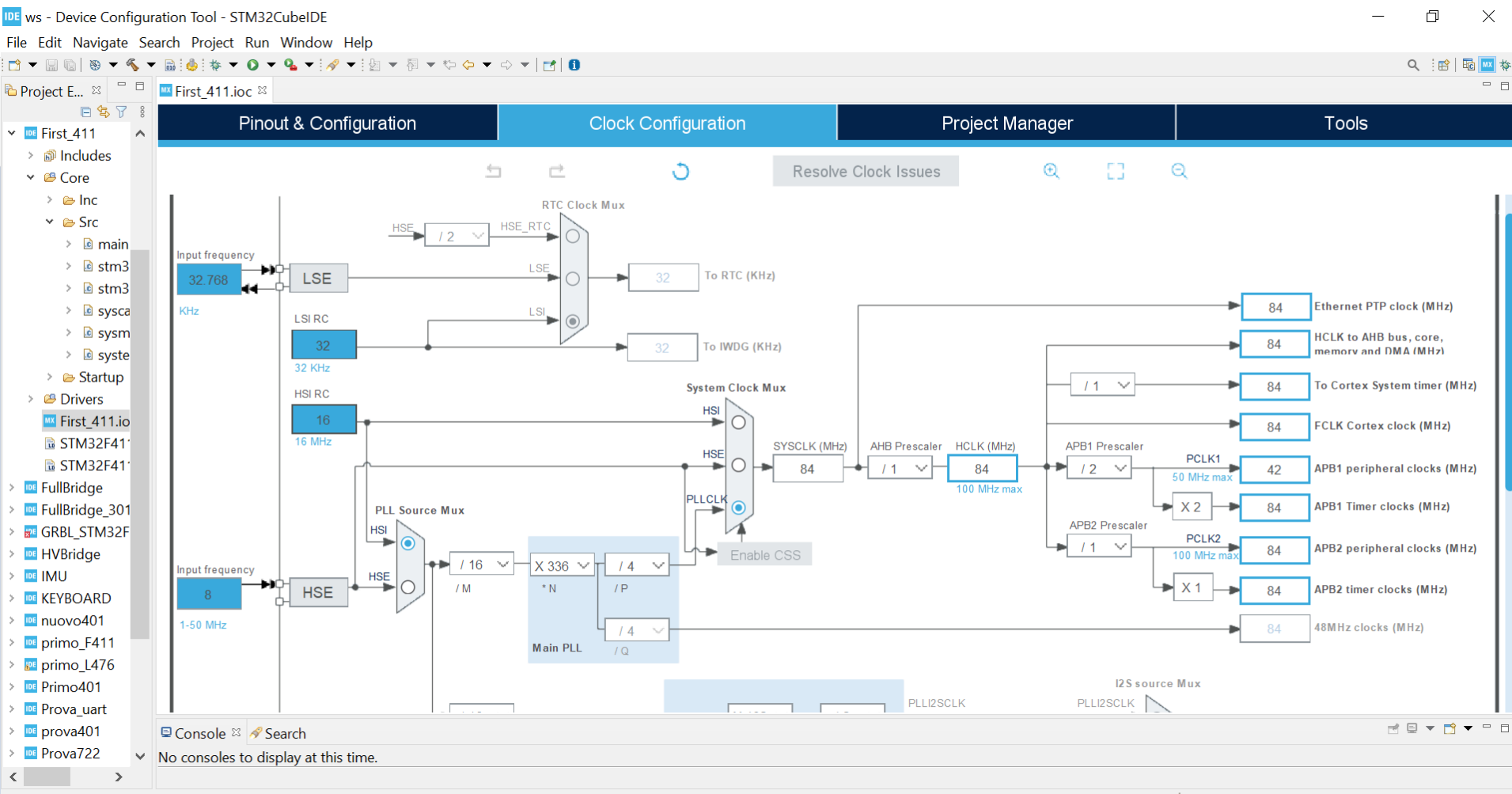
PLL2SCLK

PLL2SCLK

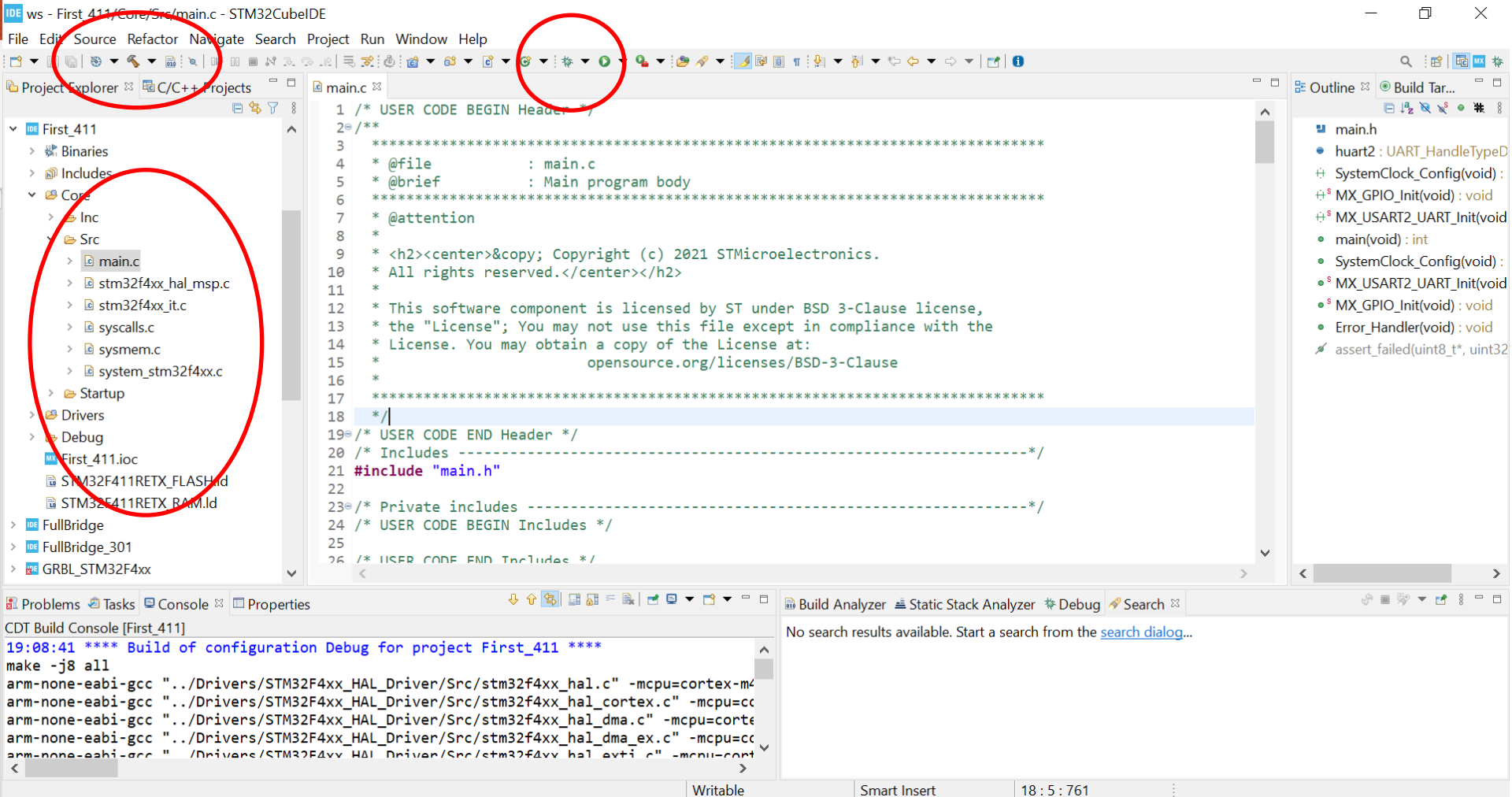
Console Search

No consoles to display at this time.

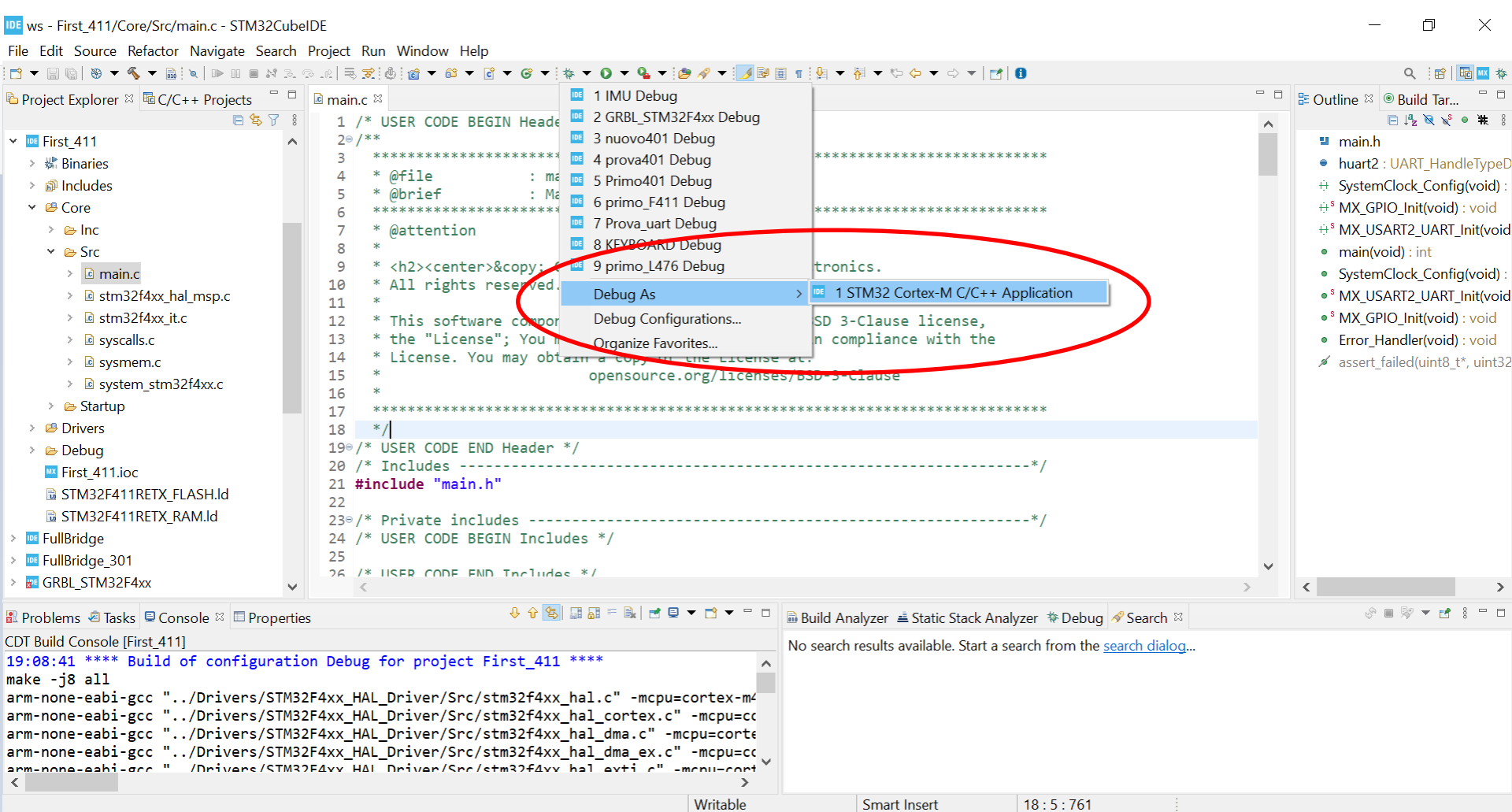
Target Configuration: clock



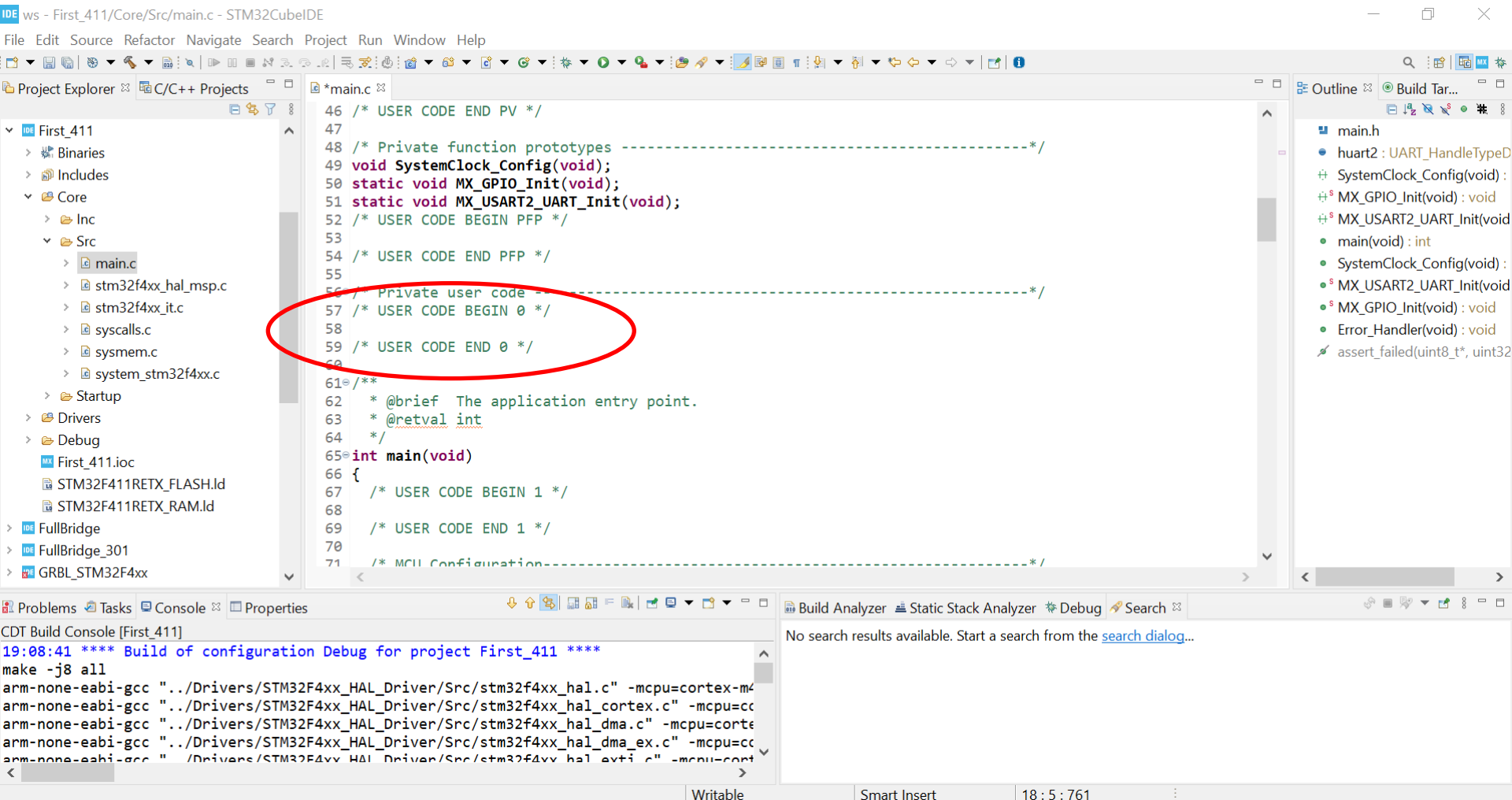
Development environment



Downloading the code and debugging



Where to write your code...



The screenshot shows the STM32CubeIDE interface. The Project Explorer on the left shows the project structure for 'First_411'. The main editor displays the 'main.c' file. A red circle highlights the 'Private user code' section, which is located between the '/* USER CODE BEGIN PFP */' and '/* USER CODE END PFP */' markers. The code in this section includes the 'main' function, which is the entry point of the application. The console at the bottom shows the build output for the project.

```

46 /* USER CODE END PV */
47
48 /* Private function prototypes -----*/
49 void SystemClock_Config(void);
50 static void MX_GPIO_Init(void);
51 static void MX_USART2_UART_Init(void);
52 /* USER CODE BEGIN PFP */
53
54 /* USER CODE END PFP */
55
56 /* Private user code -----*/
57 /* USER CODE BEGIN 0 */
58
59 /* USER CODE END 0 */
60
61 /**
62  * @brief The application entry point.
63  * @retval int
64  */
65 int main(void)
66 {
67     /* USER CODE BEGIN 1 */
68
69     /* USER CODE END 1 */
70
71     /* MCU Configuration -----*/

```

Build Console [First_411]

```

19:08:41 **** Build of configuration Debug for project First_411 ****
make -j8 all
arm-none-eabi-gcc "-./Drivers/STM32F4xx_HAL_Driver/Src/stm32f4xx_hal.c" -mcpu=cortex-m4
arm-none-eabi-gcc "-./Drivers/STM32F4xx_HAL_Driver/Src/stm32f4xx_hal_cortex.c" -mcpu=cortex-m4
arm-none-eabi-gcc "-./Drivers/STM32F4xx_HAL_Driver/Src/stm32f4xx_hal_dma.c" -mcpu=cortex-m4
arm-none-eabi-gcc "-./Drivers/STM32F4xx_HAL_Driver/Src/stm32f4xx_hal_dma_ex.c" -mcpu=cortex-m4
arm-none-eabi-gcc "-./Drivers/STM32F4xx_HAL_Driver/Src/stm32f4xx_hal_exti.c" -mcpu=cortex-m4

```